IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 10 June 2025, accepted 30 June 2025, date of publication 10 July 2025, date of current version 18 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3586498

RESEARCH ARTICLE

MulMatch-CL: A Semi-Supervised Teacher-Student Framework for Robust Crop Segmentation in UAV Imagery

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This work was supported in part by the Basic Scientific Research Project of Wenzhou under Grant N20220016, in part by the Scientific Research Fund of Zhejiang Provincial Education Department under Grant Y202353926, and in part by the Philosophy and Social Science Planning Project of Zhejiang Province of China under Grant 25NDJCO94YBM.

ABSTRACT Accurate crop segmentation using unmanned aerial vehicle (UAV) imagery is essential for efficient crop monitoring and management. While Transformer-based architectures have demonstrated exceptional performance in segmentation tasks, their application to UAV imagery remains challenging owing to limited labeled data and noisy annotations. To address these challenges, this study proposes MulMatch-CL, a novel teacher-student architecture that integrates the semi-supervised MulMatch framework with Confident Learning (CL). The proposed MulMatch component employs consistency regularization and multiple strong augmentation streams to effectively utilize unlabeled data and enhance model generalization. A teacher model is first trained on both labeled and unlabeled data to generate pixel-wise probability distributions. Confident Learning then identifies and filters noisy labels, refining the labeled dataset. The cleaned dataset, combined with the unlabeled data, is used to train a student model, resulting in a more robust segmentation performance. Experimental results on the Barley Remote Sensing Dataset show that MulMatch-CL achieves 78.42% mIoU, 89.10% pixel Acc, and 87.58% F_1 score, outperforming supervised baselines, robust learning strategies, and semi-supervised methods. Ablation studies further confirm that both Confident Learning and MulMatch independently enhance performance, improving mIoU by 2.36% and 4.28% respectively, while their integration yields a 6.08% improvement over the baseline. These results demonstrate that MulMatch-CL provides a robust solution for applying Transformer models to UAV-based crop segmentation.

INDEX TERMS Crop segmentation, UAV imagery, transformer, MulMatch, confident learning, multiple strong augmentation streams.

I. INTRODUCTION

Crop monitoring and management are essential components of smart agriculture, relying on advanced technologies to optimize crop growth [1], enable early issue detection [2], and enhance resource efficiency [3]. Among these technologies, semantic segmentation, a fundamental computer vision task that assigns crop labels to each pixel, plays a key role in agricultural applications such as weed detection [4], disease identification [5], and resource allocation [6]. The emergence of unmanned aerial vehicles (UAVs) has further expanded these capabilities by providing high-resolution aerial imagery for detailed crop analysis [7].

Deep learning has revolutionized image analysis, offering state-of-the-art solutions for semantic segmentation and related tasks [8], [9]. In particular, transformer-based architectures, such as the Swin Transformer [10], have demonstrated superior performance by capturing both local

The associate editor coordinating the review of this manuscript and approving it for publication was Alexander Kocian.

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and global features through a hierarchical design and a shifted window mechanism. These characteristics make the Swin Transformer a strong alternative to conventional convolutional neural networks (CNNs), particularly for handling the complex patterns found in crop images. Despite their success, deep learning models for UAV-based agricultural segmentation face significant challenges primarily because of the need for large-scale annotated datasets and the presence of noisy labels [11]. Pixel-wise annotation is both time-consuming and labor-intensive, particularly in remote sensing applications where high-resolution images require precise labeling [12], [13]. Additionally, label noise poses a challenge in semantic segmentation because of the spatial dependencies between pixels and the potential for errors at both fine and coarse levels [14], [15]. These factors affect model performance, necessitating strategies to mitigate the impact of label noise.

To address these challenges requires a framework capable of learning effectively from limited annotated data while being robust to noisy labels. To this end, this study proposes MulMatch-CL, a novel teacher-student algorithm that integrates the semi-supervised MulMatch framework with Confident Learning. The proposed MulMatch employs consistency regularization and multiple strong augmentation streams to fully exploit unlabeled data. Confident Learning [16], originally developed for classification tasks, is adapted for labels denoising in segmentation by estimating pixel-level confidence and constructing a pixel-wise confident joint matrix. By combining these methods, the proposed method reduces the reliance on extensive manual annotations and mitigates the impact of noisy labels in the training set.

The proposed algorithm operates in three stages. First, the teacher model is trained on both labeled and unlabeled data, generating pixel-wise probability distributions to identify annotation inconsistencies. Second, Confident Learning is applied to detect and remove noisy annotations based on the predictions of the teacher model, refining the labeled dataset. Finally, the student model is trained using the cleaned labeled data along with the unlabeled data, ensuring robust and accurate crop segmentation. MulMatch-CL has been evaluated on a UAV-based crop segmentation dataset through extensive experiments and ablation studies. The results demonstrate the robustness and adaptability of the model, setting a new benchmark for crop segmentation in scenarios with limited labeled data and noisy annotations.

To address the challenges of limited labeled data and noisy annotations in UAV-based crop segmentation, this paper makes the following key contributions:

1) This study introduces MulMatch-CL, a hybrid teacher-student architecture that integrates semi-supervised learning with Confident Learning to simultaneously address data scarcity and annotation noise in pixel-level segmentation tasks.
2) The MulMatch component utilizes consistency regularization and multiple strong augmentation streams, enabling effective exploitation of unlabeled data and enhancing model generalization.
3) This study is the first work to adapt Confident Learning into a semi-supervised framework for pixel-level segmentation in UAV imagery. Confident Learning is adapted to detect and remove noisy pixel annotations based on teacher model predictions, substantially improving training data quality and final student model performance.
4) All components are novelly integrated into a unified framework and validated through extensive experiments on the Barley Remote Sensing Dataset. Experimental results show that MulMatch-CL achieves state-of-the-art results, with a mIoU of 78.42%, pixel Acc of 89.10%, and F_1 score of 87.58%, validating the efficacy of the proposed method in crop segmentation.

II. RELATED WORK

Semantic segmentation has emerged as an essential tool for UAV-based remote sensing in agriculture. Assigning crop labels to each pixel, enables detailed crop analysis, facilitating tasks such as crop classification, disease identification, and weed detection [17]. Advances in UAV technology have made high-resolution image acquisition feasible, laying the groundwork for semantic segmentation in precision agriculture. For instance, CCTNet [18], which combines a Coupled CNN and Transformer Network, leverages UAV imagery to achieve accurate crop-type segmentation and outperforms traditional methods. Similarly, UAV-based segmentation models have shown high accuracy in separating weeds from crops [19]. This precision minimizes chemical usage, reduces cost, and enhances sustainability. Furthermore, semantic segmentation aids in identifying intra-field crop variability and supports site-specific management strategies [20].

The integration of terrain information from segmentation masks enables efficient and safe path planning for UAVs [21]. This approach facilitates the construction of navigation maps, detection of field boundaries, and adaptive focus on regions of interest, thereby enhancing both operational efficiency and agricultural outcomes. For example, interactive segmentation-based methods have been employed for agricultural field boundary detection, providing critical support for generating high-precision navigation maps for autonomous agricultural machinery [22]. Additionally, an improved ENet-based semantic segmentation model has been developed to perform row segmentation in farmland images, enabling autonomous UAV navigation and precise operations in agricultural settings [23].

A. ADVANCEMENTS IN DEEP SEMANTIC SEGMENTATION METHODS

Traditional semantic segmentation methods rely heavily on hand-crafted features. However, these approaches are limited by their dependence on feature engineering and inability to capture high-level semantic context effectively. The emergence of deep learning has significantly improved

segmentation accuracy and scalability [24]. Fully Convolutional Network (FCN) [25] is the first deep learning-based model designed for pixel-wise predictions, using convolutional layers and upsampling techniques to produce dense outputs. Subsequent CNN-based models, including U-Net [26], DeepLabv3+ [27], and PSPNet [28], introduce advanced techniques such as encoder-decoder architectures, atrous spatial pyramid pooling, and pyramid pooling modules. These innovations enhance spatial detail preservation, global context capture, and the handling of complex scenes.

The introduction of Vision Transformers (ViTs) [29] has marked a paradigm shift in computer vision, moving from convolution-dominated architectures to self-attention mechanisms. ViT processes an image as a sequence of fixed-size patches and embeds them into feature vectors processed by transformer layers. The Swin Transformer [10] incorporates a hierarchical structure with shifted windows, enabling efficient high-resolution image processing and multi-scale feature learning. Similarly, Masked Transformer for Unified Segmentation (Mask2Former) [30] employs masked attention mechanisms to handle multiple segmentation tasks within a unified framework. These transformer-based models represent a significant advancement over traditional CNN-based architectures by capturing both global context and fine-grained details.

Despite the potential of deep learning-based semantic segmentation in UAV-based agriculture, several challenges persist. These include the scarcity of annotated datasets and the presence of noisy annotations.

B. DEEP SEMI-SUPERVISED SEMANTIC SEGMENTATION

Deep semi-supervised semantic segmentation (DSSS) has emerged as a solution to address the scarcity of labeled data by exploiting large volumes of unlabeled data. This approach is particularly beneficial for segmentation tasks where obtaining labeled data is costly and labor-intensive, whereas unlabeled data are readily available. DSSS methods generally fall into two categories: consistency-based and pseudo-labeling. Consistency-based methods enforce stable model predictions under different augmentations of the same data. For instance, Cross-Consistency Training (CCT) [12] applies multiple perturbation strategies to ensure consistent outputs across various feature representations. Pseudo-labeling methods, such as self-training [31], generate pseudo-labels for unlabeled data and treat them as ground truth for further training. Hybrid methods, such as PseudoSeg [32] and UniMatch [33], integrate consistency regularization with pseudo-labeling, following the principles of FixMatch [34]. In FixMatch, high-confidence predictions of unlabeled data are treated as pseudo-labels, while consistency regularization forces the model to produce similar outputs under various transformations of the input image. However, when applied to real-world datasets such as those collected in UAV-based crop monitoring, DSSS may encounter difficulties in handling noisy annotations.

C. LEARNING WITH NOISY ANNOTATIONS

Noisy annotations are a common issue in remote sensing datasets, arising from factors such as manual annotation mistakes, automated labeling processes, and inherent ambiguity in data. A key approach to mitigating the impact of noisy annotations is the development of noise-robust loss functions. Swin-SCE [35] employs Symmetric Cross Entropy (SCE) loss to handle noisy annotations and improve performance in crop segmentation tasks. Another line of work focuses on identifying and mitigating the influence of noisy annotations through sample selection and weighting. Co-teaching [36] trains two neural networks simultaneously, with each network selecting small-loss samples for the other, exploiting the observation that correctly labeled examples often produce lower loss values during training. Regularization techniques and architectural modifications can also improve the robustness of noisy annotations. Methods such as mixup [37] augment data by blending inputs and their corresponding labels, implicitly regularizing the model to reduce its reliance on specific noisy annotations. Label refinement techniques aim to correct noisy annotations, either during preprocessing or dynamically during training [14], [38]. Confident Learning [16] has emerged as a robust framework for identifying and mitigating the impact of noisy labels. It utilizes confidence estimates to identify potential errors in labeled data, providing a cleaner dataset for model training.

III. PROPOSED METHOD

Figure 1(a) illustrates the MulMatch-CL architecture. It also shows the process of detecting noisy annotations. Both the teacher and student models are based on MulMatch, as shown in Figure 1(b). The process begins with training the teacher model with both labeled and unlabeled data, which generates probability distributions for each pixel in the labeled set. Based on these distributions, noisy annotations are identified and pruned using pixel-level label quality scores and a confident joint matrix. The student model is then trained on the cleaned labeled and unlabeled data to produce the final segmentation results.

A. SEGMENTATION NETWORK

The UperNet [39] architecture (Figure 1(c)) is adopted as the segmentation network. To leverage the strengths of vision transformers, the conventional CNN-based backbone is replaced with the Swin Transformer [10]. The input image is first divided into non-overlapping 4×4 patches, each treated as a "token" and projected into a higher-dimensional feature space using a linear layer. The Swin Transformer processes multi-level feature maps hierarchically across four stages, progressively reducing the spatial resolution while increasing the channel depth. The multi-scale features from each stage are then fed into the Feature Pyramid Network (FPN) module of UperNet to align and fuse these features across different scales using lateral and top-down connections. Finally, a decoder head applies convolutional

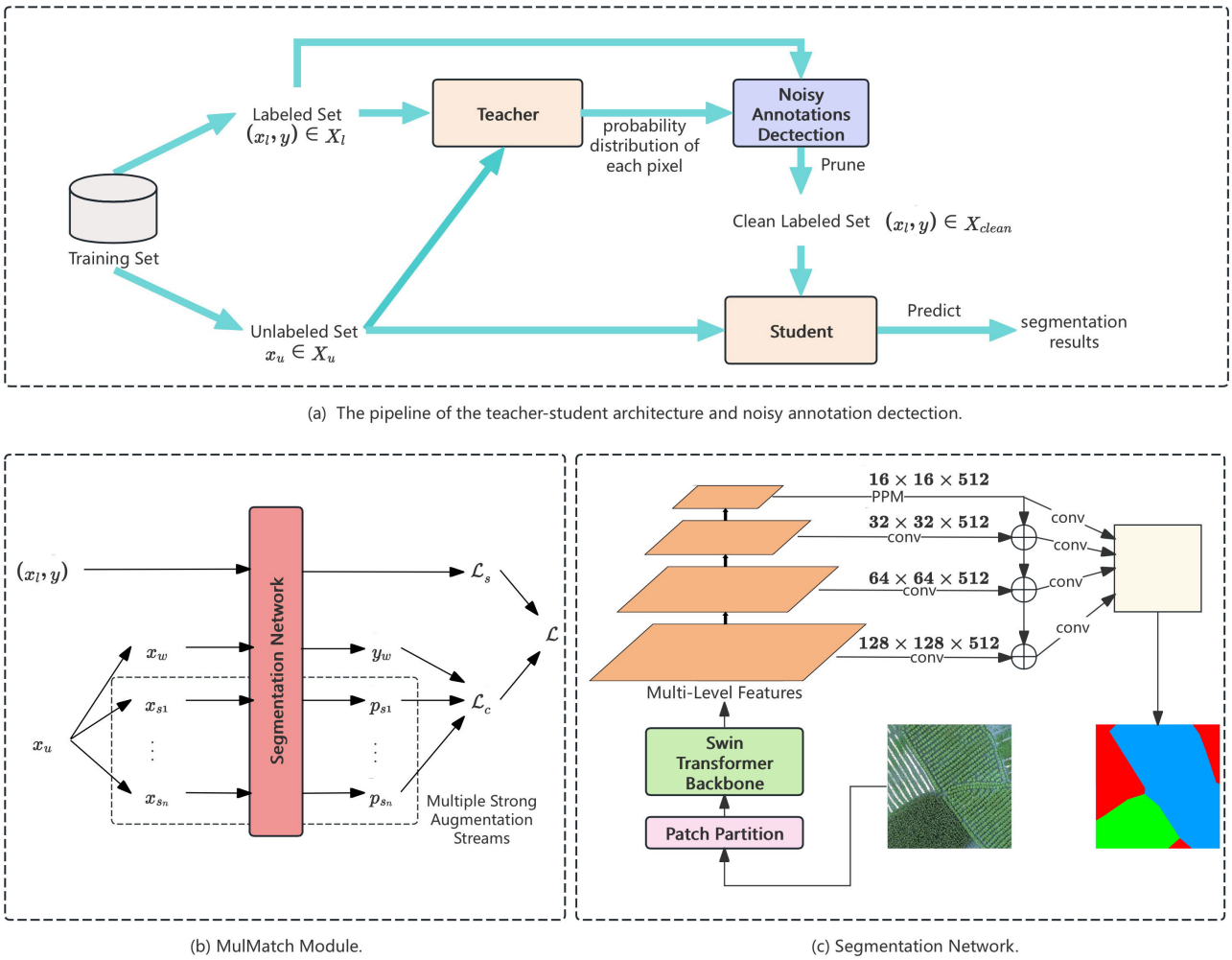

FIGURE 1. **Overall framework of the proposed method.**

operations and upsampling to produce the final semantic segmentation map at the original resolution.

B. MULMATCH MODULE

This section presents the MulMatch framework (Figure 1(c)) designed for crop segmentation tasks, including necessary modifications and theoretical justifications for each step. Weak augmentations involve simple transformations, such as horizontal flips, slight rotations, and scaling. These transformations maintain the essential features of the images while introducing minor variations, facilitating reliable pseudo-label generation. For an unlabeled image x_u, weak augmentations are first applied to obtain the image x_w. $f(x; \theta)$ denotes the probability map produced by the semantic segmentation network for image x, where θ represents the network weights. The probability map is computed as $p_w = f(x_w; \theta)$ and generate the corresponding pseudo-label map $y_w = \text{argmax}(f(x_w; \theta))$. These pseudo-labels can be correct or incorrect. When the model produces a large number of incorrect labels, it negatively impacts the training results. To improve pseudo-label reliability, a confidence threshold τ is applied to the model's output predictions y_w, retaining only those with high-confidence (above a certain threshold τ) predictions, while discarding those samples that do not meet the criteria.

Strong augmentations introduce significant variations to the input images, including color jittering, Gaussian blur, and elastic transformations. These augmentations challenge the model to maintain consistency across diverse transformations, thereby improving its robustness. Next, strong augmentations are applied to the same unlabeled image x to produce its augmented version x_s. The same semantic segmentation network with weights θ is then used to predict the segmentation probability map $p_s = f(x_s; \theta)$. Consistency loss $\mathcal{L}_c$ enforces the agreement between x_s and y_w, which is computed by the cross-entropy loss $\text{H}(p, q)$:

$$\mathcal{L}_c = \frac{1}{|\mathcal{D}^u|} \sum_{x_u \in \mathcal{D}^u} \mathbb{1}(\max(p_w) \geq \tau) \text{H}(y_w, f(x_s; \theta)) \quad (1)$$

where $\mathcal{D}^u$ is the batch of unlabeled images, x_u represents the unlabeled image, and x_s denotes its corresponding strongly augmented version.

The supervised loss $\mathcal{L}_s$ is computed on labeled data (x_l, y) using the cross-entropy loss $\mathrm{H}(p, q)$:

$$\mathcal{L}_s = \frac{1}{|\mathcal{D}^l|} \sum_{(x_l, y) \in \mathcal{D}^l} \mathrm{H}(y, f(x_l; \theta)) \tag{2}$$

The loss function of the proposed segmentation network $\mathcal{L}$ consists of a supervised loss $\mathcal{L}_s$ and a consistency loss $\mathcal{L}_c$:

$$\mathcal{L} = \mathcal{L}_s + \lambda \mathcal{L}_c \tag{3}$$

where λ is a hyperparameter that balances the influence of the consistency loss.

1) MULTIPLE STRONG AUGMENTATION STREAMS

Data augmentation is particularly useful in machine learning, especially for tasks such as semantic segmentation, where models must be robust to various transformations and noise. Inspired by the CCT [12] method, this study introduces multiple strong augmentation streams into the MulMatch framework. This approach ensures that the model learns to be consistent across a broader range of transformations, leading to better generalization.

As shown in Figure 1(b), the proposed MulMatch method employs multiple strong augmentation streams instead of a single stream. By applying diverse strong augmentations, the model is exposed to a wider range of variations, encouraging it to learn invariant features and maintain consistent predictions across different augmentations. Given an unlabeled image x_u, different strong augmentations are applied to produce multiple strongly augmented images denoted as $(x_{s_1}, x_{s_2}, \ldots, x_{s_n})$. The corresponding probability maps, $(p_{s_1}, p_{s_2}, \ldots, p_{s_n})$, are obtained from the same segmentation network $f(x; \theta)$, where θ represents their weights. The consistency loss $\mathcal{L}_c$ with multiple strong augmentations is defined as:

$$\mathcal{L}_c = \frac{1}{|\mathcal{D}^u|} \frac{1}{n_s} \sum_{x_u \in \mathcal{D}^u} \sum_{i=1}^{n_s} \mathbb{1}(\max(p_w) \geq \tau) \mathrm{H}(y_w, f(x_{s_i}; \theta)) \tag{4}$$

where n_s denotes the number of different strong augmentation streams.

2) CutMix IN STRONG AUGMENTATION

CutMix [40], originally designed for image classification, enhances model robustness by blending patches from different images along with their corresponding labels. In this study, the method is adapted to modify the augmentation process, particularly during strong augmentations. It does so by cutting a patch from one image and pasting it onto another, effectively combining the information from multiple samples. When combined with MulMatch, CutMix improves performance by increasing sample diversity and providing stronger perturbations.

In MulMatch, strong augmentation regularizes the model by enforcing consistency under different input perturbations. CutMix can be used as a form of strong augmentation by combining the regions of two images, encouraging the model to generalize across partial inputs. In a semi-supervised MulMatch setting, an unlabeled image x is selected. A random patch is cut from x and replaced with a patch from another image x'. CutMix operation is applied during the strong augmentation phase, and the resulting mixed image x_m is defined as:

$$x_m = M \odot x + (1 - M) \odot x' \tag{5}$$

where M denotes the binary mask, and $\odot$ represents the element-wise product.

The mixed pseudo-labels y_m is denoted as:

$$y_m = M \odot y + (1 - M) \odot y' \tag{6}$$

where y and y' are the pseudo-labels of x and x', respectively.

C. NOISY ANNOTATION DECTECTION

Label noise poses significant challenges in semantic segmentation. Label noise refers to incorrect, ambiguous, or inconsistent annotations, which can severely impact the performance of segmentation models. Meanwhile, each pixel needs to be labeled correctly, which significantly increases the complexity of identifying label noise compared to image classification tasks. Given these challenges, Confident Learning is adapted to enhance pixel-wise predictions in semi-supervised semantic segmentation tasks. Originally developed for supervised learning, Confident Learning focuses on identifying mislabeled examples in training data, thereby enhancing model performance. Building on this framework, this study proposes MulMatch-CL, which integrates Confident Learning with semi-supervised learning to address label noise effectively.

Specifically, a teacher model *seg* is trained using the MulMatch methods described in Sections III-A and III-B. Given a $h \times w$ pixel image x with associated label $y \in \mathbb{Z}^{h \times w}$, the model generates a probability distribution (via softmax) $seg_x = seg(x) \in \mathbb{R}^{h \times w \times C}$ over C classes for each pixel. Here, $seg_x[i][j][c]$ represents the predicted value of the pixel at position (i, j) belonging to class c. The goal of confident learning is to identify noisy or incorrect labels in a dataset by comparing the observed labels to the predicted class probabilities. These probabilities are more reliable when estimated from a model that is not overfitted to the training data. Thus, the dataset is divided into K-folds. The predicted class probabilities p_k from each fold are collected and aggregated. At the end of K-folds cross-validation, each instance in the dataset obtains out-of-sample class probability predictions.

1) ESTIMATE THE PIXEL-LEVEL CONFIDENCE

By analyzing the probability distribution seg_x and associated label y, it is possible to compute a label quality score for each pixel at position (i, j). For each pixel, its label quality score $s_x(i, j)$ is defined as the softmax probability of the predicted class:

$$s_x(i, j) = seg_x[i][j][y_{ij}] \tag{7}$$

where $y_{ij} \in \{0, \ldots, C-1\}$ denotes the class label. Pixels where the predicted class has high confidence but does not match the ground truth, are likely candidates for noisy labels.

For each pixel (i, j), the confidence score is calculated as the maximum probability across classes:

$$\mathrm{Conf}_x(i,j) = \max_c seg_x[i][j][c] \tag{8}$$

Pixels with confidence scores exceeding a predefined threshold τ are considered high-confidence predictions:

$$\mathrm{Conf}_x(i,j) \geq \tau_x(i,j) \tag{9}$$

Only pixels that satisfy this condition are used to build the confident joint matrix. For each high-confidence pixel (i, j), the predicted class is assigned as:

$$\hat{c}_x(i,j) = \arg\max seg_x[i][j] \tag{10}$$

where $\hat{c}_x(i,j) \in \{0, \ldots, C-1\}$. This provides the predicted class of the model for each confident pixel. The threshold $\tau_x(i,j)$ for pixel (i, j) of image x is denoted as:

$$\begin{aligned}\tau_x(i,j) &= \tau_{\hat{c}_x(i,j)} \\ &= \frac{\sum_{(x',y')\in X}\left(\sum_{\{(a,b):y'_{ab}=\hat{c}(i,j)\}} seg_{x'}[a][b][\hat{c}_x(i,j)]\right)}{\sum_{(x',y')\in X}\left|\{(a,b): y'_{ab} = \hat{c}_x(i,j)\}\right|}\end{aligned} \tag{11}$$

where $\tau_{\hat{c}_x(i,j)} \in \{\tau_0, \ldots, \tau_{C-1}\}$ and $X = (x, y)^n$ denotes the set of n images with associated labels.

2) BUILD THE CONFIDENT JOINT MATRIX FOR PIXELS

The confident joint matrix captures the relationship between the predicted and true labels, particularly focusing on high-confidence predictions. Adapting this concept to semantic segmentation involves calculating a similar matrix at the pixel level. The confident joint matrix $J \in \mathbb{R}^{C\times C}$ is defined such that each entry $J[c_i][c_j]$ represents how many pixels with true label c_i are predicted as class c_j. The steps to build the matrix are as follows:

Algorithm 1: Steps to Build the Confident Joint Matrix

Input: The matrix $J \in \mathbb{R}^{C\times C}$;
Output: The confident joint matrix $J \in \mathbb{R}^{C\times C}$;
Step 1: Initialize the Confident Joint Matrix
Create a $C\times C$ matrix J, initialized with zeros;
Step 2: Iterate over all pixels
for all pixels **do**
  if the confidence score $\mathrm{Conf}_x(i,j) \geq \tau_x(i,j)$ **then**
    Increment the corresponding entry in the matrix:

$$J[y_{ij}][\hat{c}_x(i,j)] = J[y_{ij}][\hat{c}_x(i,j)] + 1$$

    $\hat{c}_x(i,j)$ is the predicted class;
    y_{ij} is the ground truth (true) class for that pixel;
  end if
end for

This counts how often each true class y_{ij} is predicted as class $\hat{c}_x(i,j)$ across all the high-confidence pixels. To interpret the matrix in terms of probabilities, the matrix J is normalized. This normalization step provides the estimated joint probability distribution J' between the predicted class j' and the true label i':

$$\begin{aligned}&J'[c_i][c_j] \\ &= \frac{\frac{J[c_i][c_j]}{\sum_{b=1}^{C} J[c_i][b]} \cdot \left|\{(x,y,i,j) : (x,y) \in X,\ y_{ij} = c_i\}\right|}{\sum_{a=1,b=1}^{C}\left(\frac{J[a][b]}{\sum_{b'=1}^{C} J[a][b']} \cdot \left|\{(x,y,i,j) : (x,y) \in X,\ y_{ij} = a\}\right|\right)}\end{aligned} \tag{12}$$

3) DETECT AND PRUNE NOISY PIXELS

Noisy pixels can be identified by analyzing the joint probability distribution $J'[y_{ij}][\hat{c}_x(i,j)]$ and the label quality score $s_x(i,j)$. Pixels with a lower label quality score $s_x(i,j)$ are more likely to be noisy. For each class $c \in \{0, \ldots, C-1\}$, select

$$n \cdot h \cdot w \cdot \sum_{c_j=1, c_j\neq c}^{C} (J'[c][c_j])$$

pixels with the lowest label quality score $s_x(i,j)$ as the noisy pixel sample set X_{noisy}. Noisy pixels are subsequently excluded from the entire training process. This can be achieved by setting the loss to zero for noisy pixels:

$$\mathcal{L}_{filtered} = \sum_{(x,y)\in X_{clean}} (\mathcal{L}(seg(x), y)) \tag{13}$$

Only pixels that are not marked as noisy are considered during the training of the Student model.

IV. EXPERIMENTS

A. EXPERIMENTAL SETTINGS

1) IMPLEMENTATION DETAILS

The confidence threshold τ is set to 0.95. The influence weight λ for the consistency loss is set to 1, and the number of different strong augmentation streams n_s is set to 2. For Transformer-based backbones, AdamW is used with a learning rate of 6e-5 and weight decay of 0.01. A polynomial learning rate schedule (Poly) is employed, defined as:

$$lr = lr_{init} \times \left(1 - \frac{iter}{max_iter}\right)^{power}$$

where the *power* is set to 1.0. To improve training stability during the initial phase, a linear warmup is applied during the first 10 epochs. For CNN-based backbones, Stochastic Gradient Descent (SGD) is used with a learning rate of 0.01, momentum of 0.9, weight decay of 5e-4, and the power of the Poly learning rate schedule is set to 0.9. The model is trained for 100 epochs with a batch size of 16. All models are trained with the same total number of epochs to ensure a fair comparison.

To enhance model generalization and improve robustness, a series of data augmentation techniques are applied during training. Specifically, input images are randomly resized within a ratio range of (0.5, 2.0) to introduce scale variations. Fixed-dimension crops are randomly extracted from the resized images. Horizontal flipping is applied with

a 50% probability. To simulate diverse lighting conditions, brightness, contrast, saturation, and hue adjustments are randomly applied.

2) DATASETS

Barley Remote Sensing Dataset [41]: The dataset used in this study is collected from barley fields in Xingren City, Guizhou Province, China. The selection of this region is attributed to its agricultural significance as a representative barley-producing area, as well as its complex geographic and climatic conditions. Xingren's landscape features undulating hills, fragmented plots, and varied elevations, along with fluctuating light conditions and seasonal variability in vegetation growth. These characteristics present realistic challenges for aerial image acquisition, segmentation, and model generalization. The dataset includes both raw images and their corresponding annotations, where pixel values denote specific crop types: 'Background' (0), 'Tobacco' (1), 'Corn' (2), and 'Barley' (3).

The dataset is illustrated in Figure 2, which comprises four very large, high-resolution images, with dimensions of $(44,647 \times 32,881)$ pixels, $(55,128 \times 49,447)$ pixels, $(44,343 \times 33,603)$ pixels, and $(18,576 \times 68,363)$ pixels, respectively. These image sizes highlight the need for efficient patch-based processing or tiling strategies, which are critical for crop segmentation tasks. First, the images are resized to half of their original width and height. Next, a cropping approach using a sliding window with a patch size of (512×512) pixels and a stride of 256 pixels. After cropping, 12,803 images are obtained.

To assess the crop segmentation performance under limited labeled data, a subset of the images is selected as labeled data for the experiment, while the remaining images serve as additional unlabeled data. Specifically, the training set consists of 917 labeled images, the validation set includes 512 labeled images, and the test set contains 1,108 labeled images. Figure 3 presents the results of the cropping process.

3) EVALUATION METRICS

Model performance is evaluated using standard segmentation metrics, including mean Intersection over Union (mIoU), pixel accuracy (pixel Acc), precision, recall, and the F_1 score. To assess computational complexity, the number of parameters (Param.) is reported to indicate memory requirements, floating point operations per second (FLOPs) are measured to reflect computational cost, and frames per second (FPS) is used to quantify inference speed.

B. ABLATION STUDY

To understand the contributions of each component in the proposed methodology, we analyze the results of the ablation studies presented in Tables 1-4. Section IV-B1 evaluates the incremental impact of Confident Learning and MulMatch on the baseline UperNet model with a Swin-T backbone. Section IV-B2 examines the effect of multiple strong augmentation streams, while Section IV-B3 compares the performance of different backbone architectures. Finally, Section IV-B4 evaluates the effectiveness of incorporating CutMix.

1) IMPACT OF MULMATCH AND CONFIDENT LEARNING

As shown in Table 1, the baseline, utilizing the UperNet architecture with a Swin-T backbone, achieves a mIoU of 72.34%, a pixel Acc of 85.39%, and a F_1 score of 83.35%. These results provide an initial comparison against the improvements brought by each component of the proposed methods. Integrating Confident Learning with the baseline improves mIoU by 2.36%, from 72.34% to 74.70%, while pixel Acc improves by 1.91%. These gains suggest that Confident Learning effectively mitigates label noise and enhances model reliability. Incorporating MulMatch with the baseline yields an even greater improvement, with mIoU increasing by 4.28% to 76.62% and pixel Acc increasing by 3.12%.

The combination of MulMatch and Confident Learning achieves the best overall performance, with mIoU reaching 78.42% and pixel Acc improving to 89.10%. While both MulMatch and Confident Learning independently enhance performance, their combination achieves cumulative improvements, suggesting that the two methods address different challenges in crop segmentation. MulMatch leverages unlabeled data, while Confident Learning refines the labeled data, creating a clean and informative dataset for the student model. As a result, the model can generalize better and achieve more precise crop segmentation results.

TABLE 1. Ablation study on the impact of Confident Learning and MulMatch.

Method	mIoU	pixel Acc	F_1
Baseline	72.34	85.39	83.35
Baseline + CL	74.70	87.30	85.13
Baseline + MulMatch	76.62	88.51	86.45
Baseline + MulMatch + CL	78.42	89.10	87.58

2) IMPACT OF MULTIPLE STRONG AUGMENTATION STREAMS

The results in Table 2 highlight the performance gains achieved by incorporating multiple strong augmentation streams. As a baseline, FixMatch [34] with single augmentation streams achieves 75.47% mIoU, 87.23% pixel Acc, and 85.52% F_1 score. The proposed method, even with MulMatch, surpasses FixMatch across all metrics, yielding a mIoU of 76.62%, a pixel Acc of 88.51%, and a F_1 score of 86.45%. This improvement validates the effectiveness of multiple augmentation streams. The proposed method achieves peak performance when combined with multiple strong augmentation streams. This configuration achieves a mIoU of 78.42% and a pixel Acc of 89.10%, corresponding to improvements of 1.24% mIoU, and 1.08% pixel Acc over the proposed method with single augmentation streams.

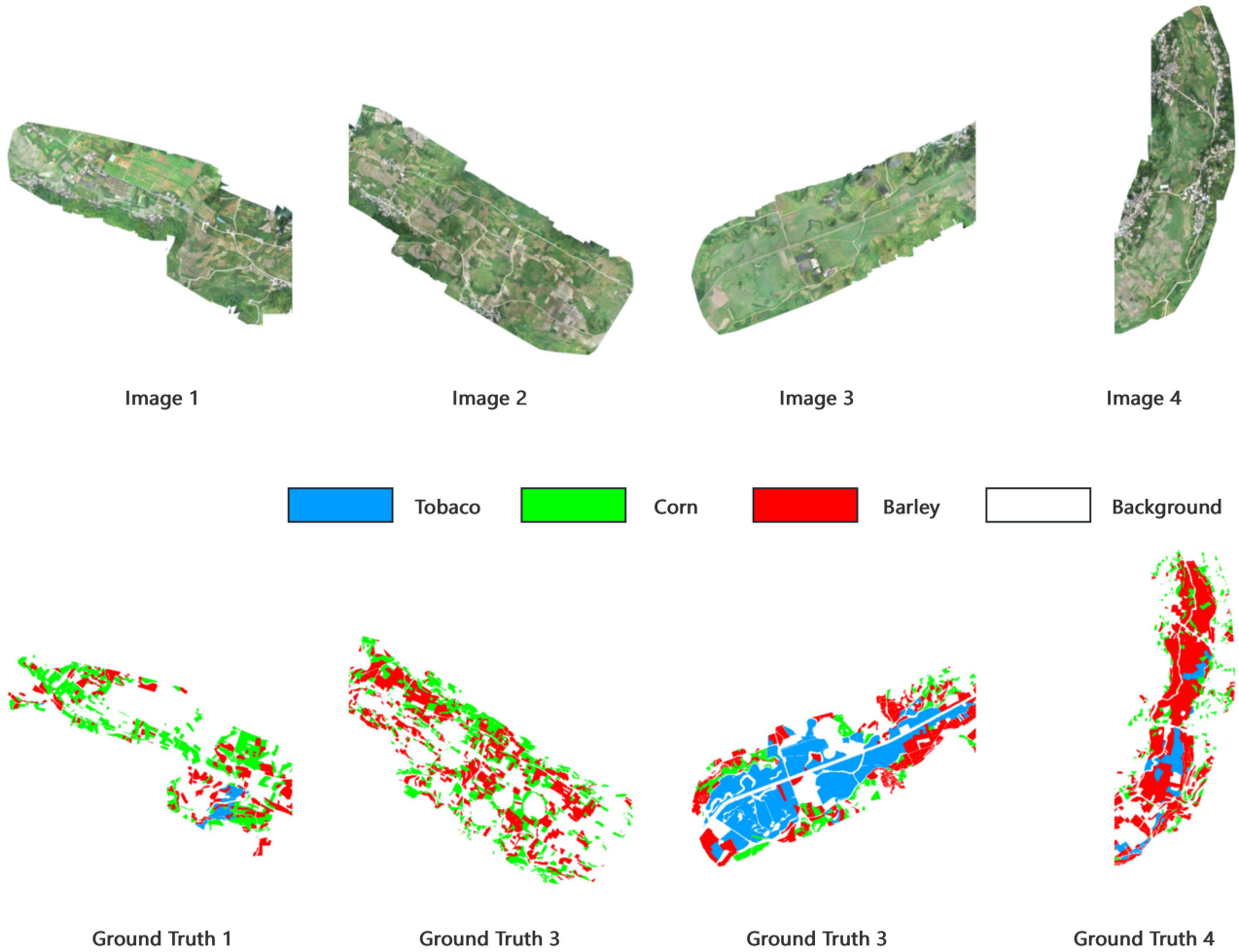

FIGURE 2. A visualization overview of the barley remote sensing dataset. Blue, green, yellow, and white represent tobacco, corn, barley, and backbone, respectively.

TABLE 2. Ablation study on the impact of multiple strong augmentation streams.

method	mIoU	pixel Acc	F_1
FixMatch	75.47	87.23	85.52
MulMatch	76.62	88.51	86.45
MulMatch-CL w/ single streams	77.18	88.02	86.65
MulMatch-CL w/ multiple streams	78.42	89.10	87.58

3) IMPACT OF BACKBONE

To assess the impact of different backbone architectures on MulMatch-CL, we compare four commonly used backbones: ResNet-50 [42], ConvNeXt-T [43], DeiT-S [44], and Swin-T [10]. The results in Table 3 demonstrate that the choice of backbone significantly influences the crop segmentation performance. Among the tested backbones, Swin-T achieves the best performance, with a mIoU of 78.42%, a pixel Acc of 89.10%, and a F_1 score of 87.58%, while maintaining a balanced computational profile. Its hierarchical structure and window-based self-attention allow effective context modeling with competitive runtime. DeiT-S, a vision transformer-based backbone, achieves 76.42% mIoU but incurs higher FLOPs and slower speed. ConvNeXt-T, adapting the simplicity and efficiency of CNNs while incorporating several design principles inspired by Vision Transformers, exhibits a balanced performance with a mIoU of 75.98%. Its performance is slightly lower than DeiT-S and Swin-T but surpasses that of ResNet-50, indicating the effectiveness of enhanced convolutional approaches in crop segmentation tasks. ResNet-50, a widely used baseline in semantic segmentation, achieves 72.11% mIoU with solid inference speed but lower accuracy compared to more modern backbones. This result reflects its limitations in crop segmentation tasks compared with transformer-based or modern CNN backbones.

This ablation study highlights the critical role of selecting a robust backbone architecture within the proposed teacher-student framework, particularly when integrating MulMatch and Confident Learning. Swin-T is the optimal

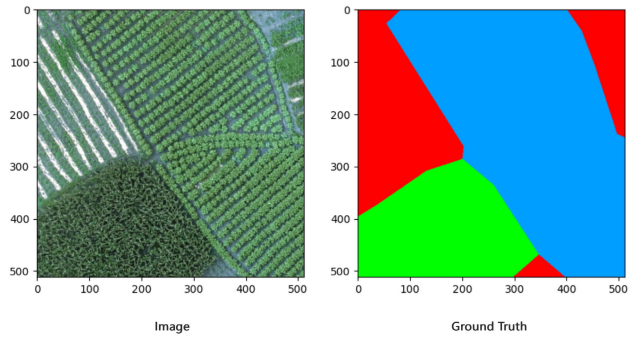

FIGURE 3. The results of applying the sliding window cropping approach to the images.

choice, leveraging its hierarchical transformer design to achieve superior segmentation performance. These results not only validate the proposed architectural decisions but also underscore the potential of transformer-based backbones in advancing crop segmentation techniques for UAV-based remote sensing applications.

TABLE 3. Ablation study on the impact of different backbone architectures.

Backbone	mIoU	pixel Acc	F_1	FLOPs	Param.	FPS
ResNet-50	72.11	84.77	83.06	237G	64.04M	21.91
ConvNeXt-T	75.98	87.85	85.92	234G	59.24M	20.86
DeiT-S	76.42	88.21	86.19	274G	57.00M	18.48
Swin-T	78.42	89.10	87.58	236G	58.94M	20.73

4) IMPACT OF CutMix

To assess the impact of CutMix augmentation on the proposed semi-supervised segmentation framework, we perform an experiment to compare the performance of the proposed method with and without CutMix. The results indicate that incorporating CutMix improves the segmentation performance, yielding a 1.35% improvement in mIoU and a 0.66% improvement in pixel Acc. This indicates that CutMix improves the generalization of the model by introducing synthetic mixed samples, which encourages better feature learning and robustness.

TABLE 4. Ablation study on the impact of CutMix.

Method	mIoU	pixel Acc	F_1
MulMatch-CL w/o CutMix	77.07	88.44	86.69
MulMatch-CL w/ CutMix	78.42	89.10	87.58

C. EXPERIMENT RESULTS

We further evaluate the performance of the proposed MulMatch-CL method against several state-of-the-art approaches. The comparison includes methods for robust learning with noisy annotations, deep semi-supervised semantic segmentation, traditional convolutional architectures, and recent transformer-based models. The goal of the experiment is to evaluate the effectiveness of MulMatch-CL for crop segmentation, particularly in the context of agricultural UAV remote sensing imagery where acquiring labeled data is challenging, and imagery is often prone to label inconsistencies.

1) COMPARISON WITH ROBUST LEARNING AND SEMI-SUPERVISED SEMANTIC SEGMENTATION METHODS

As shown in Table 5, MulMatch-CL is compared with several state-of-the-art approaches. The compared methods include (1) techniques for robust learning with noisy annotations, including symmetric cross-entropy loss (SCE) [35] and Confident Learning (CL) [16]; and (2) deep semi-supervised semantic segmentation methods, including mean teacher (MT) [45], cross-consistency training (CCT) [12], self-training (ST) [31], and FixMatch [34]. All the methods are implemented using the UperNet architecture with a Swin-T backbone.

MulMatch-CL achieves the highest segmentation performance, with a mIoU of 78.42% and pixel Acc of 89.10%, outperforming all other methods by a significant margin. The proposed method outperforms the strongest semi-supervised baseline, FixMatch, by 2.95% in mIoU and 2.06% in F_1 score. This gain highlights the synergistic benefit of combining MulMatch's multi-branch consistency with the noise-aware filtering of Confident Learning. Notably, while FixMatch achieves high precision, it suffers from slightly lower recall. In contrast, MulMatch-CL maintains a better balance between precision and recall, leading to a superior F_1 score of 87.58%. Specifically, MulMatch-CL shows superior performance across all crop categories, particularly for Barley (65.93% IoU) and Corn (72.18% Iou), where it significantly surpasses other approaches. Compared to the baseline, the proposed approach improves mIoU by 6.08%, pixel Acc by 3.71%, and F_1 score by 4.23%. The high pixel Acc suggests fewer misclassified pixels, while the highest mIoU reflects the method's ability to avoid overfitting to dominant classes, which is an important feature for datasets with class imbalance. The mIoU is also a robust measure for segmentation tasks as it reflects the ability to distinguish between classes. As shown in Figure 4, this difference is particularly notable in segmentation tasks, where a higher mIoU indicates more precise object delineation and spatial coherence, essential for accurately mapping agricultural fields. This is crucial in agricultural remote sensing, where variability in crop growth stages or other land cover elements might require consistent performance across all classes.

Among the comparison methods, FixMatch performs the best in the semi-supervised setting, achieving 75.47% mIoU, 87.23% pixel Acc, and 85.85% F_1 score. This indicates that consistency regularization with strong augmentations is beneficial for learning from unlabeled data. Self-training also shows a competitive performance with 74.90% mIoU, as it benefits from the iterative refinement of pseudo-labels. The MT method performs similarly to self-training, with a mIoU of 74.18% and a pixel Acc of 86.32%.

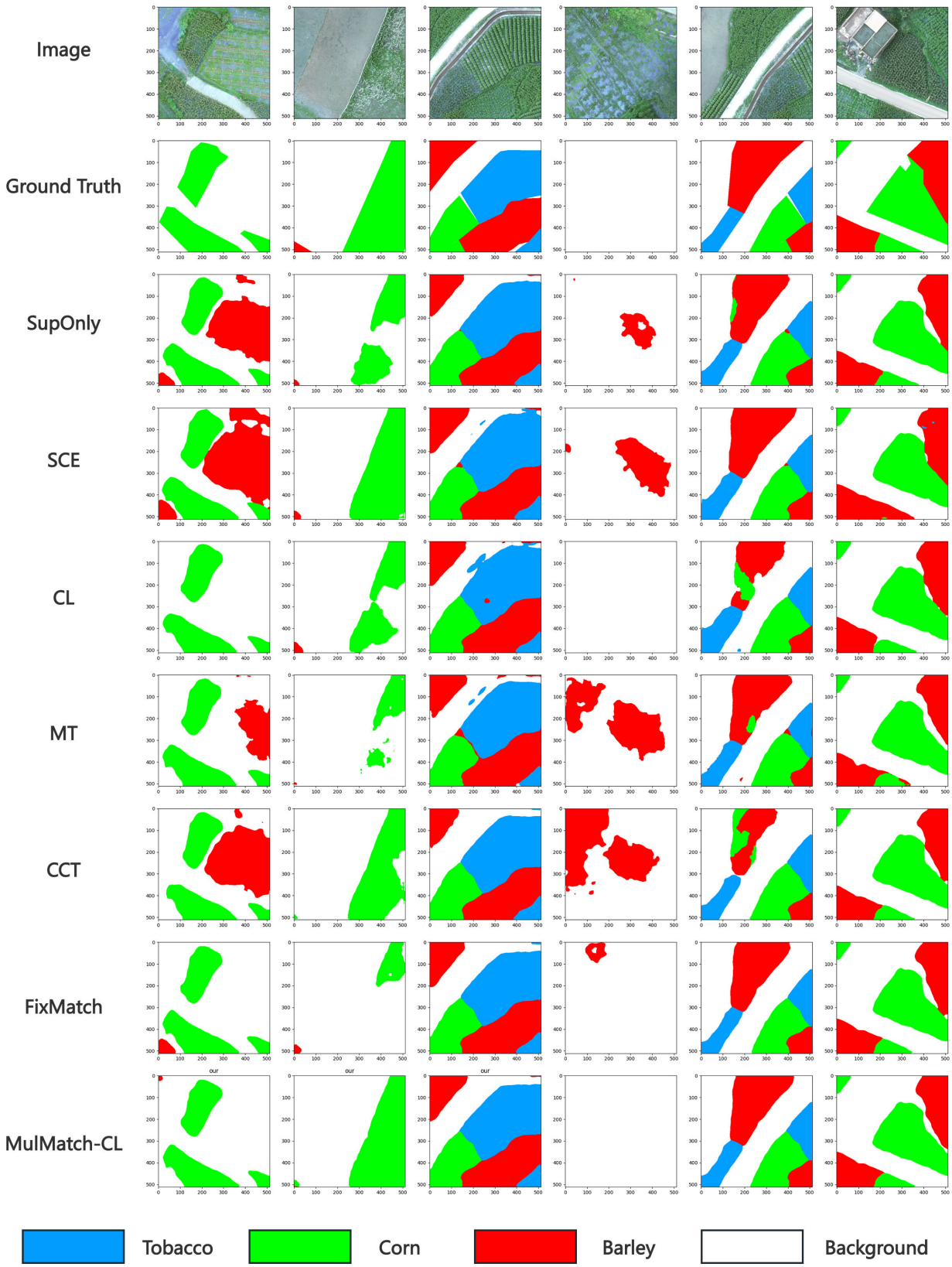

FIGURE 4. Visualization of the results.

TABLE 5. Quantitative results comparing the proposed method with robust learning techniques for noisy annotations and deep semi-supervised semantic segmentation approaches. All methods are built on the UperNet architecture with a Swin-T backbone.

Method	IoU per class				mIoU	pixel Acc	Precision	Recall	F_1
	Tobacco	Corn	Barley	Other					
SupOnly	85.12	66.40	55.23	82.63	72.34	85.39	84.64	82.64	83.35
SCE	85.82	68.54	53.22	83.78	72.84	85.62	84.06	84.14	83.59
CL	83.98	68.75	61.18	84.91	74.70	87.30	85.73	84.89	85.13
MT	89.06	70.60	54.16	82.90	74.18	86.32	86.78	82.67	84.48
CCT	88.09	69.79	53.06	82.01	73.24	85.41	84.30	83.53	83.83
ST	87.79	67.92	59.62	84.27	74.90	86.90	85.92	85.20	85.14
FixMatch	**89.15**	65.78	62.38	84.58	75.47	87.23	88.24	83.75	85.52
MulMatch-CL	88.84	**72.18**	**65.93**	**86.74**	**78.42**	**89.10**	**88.83**	**86.51**	**87.58**

TABLE 6. Quantitative results for different architectures and backbones on the barley remote sensing dataset test set.

Method	Backbone	IoU per class				mIoU	pixel Acc	Precision	Recall	F_1	FLOPs	Param.	FPS
		Tobacco	Corn	Barley	Other								
FCN	ResNet-50	81.76	66.56	47.83	79.52	68.92	82.93	82.29	79.79	80.80	198G	47.13M	21.14
PSPNet	ResNet-50	84.71	67.16	47.87	79.43	69.79	83.26	81.85	81.09	81.34	179G	46.60M	20.75
DeepLabV3+	ResNet-50	83.73	64.78	47.56	78.54	68.65	82.18	81.33	80.42	80.55	176G	41.22M	18.43
UperNet	ResNet-50	82.52	67.62	49.38	80.89	70.10	83.92	82.89	80.68	81.67	237G	64.04M	21.77
CCNet	ResNet-50	86.72	61.02	48.08	76.87	68.17	81.42	81.43	79.76	80.14	201G	47.45M	19.82
OCRNet	HRNet-w48	83.55	66.26	49.89	80.73	70.11	83.63	82.23	81.93	81.66	162G	70.35M	12.63
SegFormer	MiT-B4	82.40	66.52	50.18	80.88	69.99	84.08	83.93	79.85	81.62	59G	61.37M	15.64
Mask2Former	Swin-T	78.02	66.85	54.08	82.22	70.29	85.25	83.27	81.50	82.06	69G	47.40M	16.27
UperNet	Swin-T	85.12	66.40	55.23	82.63	72.34	85.39	84.64	82.64	83.35	236G	58.94M	20.69
MulMatch-CL	**Swin-T**	**88.84**	**72.18**	**65.93**	**86.74**	**78.42**	**89.10**	**88.83**	**86.51**	**87.58**	236G	58.94M	20.73

The CCT method, which enforces consistency between predictions under different augmentations, performs the worst among semi-supervised approaches. This suggests that the consistency regularization alone does not significantly enhance performance for this task. The introduction of SCE loss yields a slight increase in mIoU. This suggests that although SCE may handle noisy labels better than the baseline, it does not lead to significant improvements in segmentation quality. In contrast, Confident Learning, which corrects noisy labels based on the confidence of the model, shows notable improvements over the baseline, with a 2.36% increase in mIoU, a 1.91% increase in pixel Acc, and a 1.78% increase in F_1. This demonstrates that leveraging label correction can be beneficial for segmentation tasks involving noisy annotations.

Overall, the results demonstrate the robustness and practical applicability of MulMatch-CL for precise segmentation in challenging agricultural scenarios by combining noisy label correction and consistency regularization to enhance performance. By improving the data quality and efficiently leveraging limited labeled data, the proposed method achieves superior segmentation performance, providing a robust solution for agricultural remote sensing tasks.

2) COMPARISON WITH DIFFERENT SEMANTIC SEGMENTATION ARCHITECTURES AND BACKBONES ON THE BARLEY REMOTE SENSING DATASET TEST SET

Table 6 presents a detailed comparison of the proposed method with different architectures and backbones. The compared methods include (1) CNN-Based methods: FCN [25], PSPNet [28], DeepLabV3+ [27], UperNet (ResNet-50) [39], CCNet [46], OCRNet [47]; and (2) Transformer-Based methods: SegFormer [48], Mask2Former [30], UperNet (Swin-T) [10]. CNN-based methods represent the traditional paradigm in semantic segmentation, leveraging convolutional operations for local feature extraction and spatial hierarchy learning. Transformer-based architectures have emerged as robust alternatives to CNNs, leveraging self-attention mechanisms for global context modeling and long-range dependencies.

The proposed method based on UperNet with a Swin-T backbone achieves the highest scores across all metrics. As shown in Table 6, it significantly outperforms all other methods with a 6.08% absolute gain in mIoU over the standard UperNet-Swin-T baseline, while maintaining the same computational cost (same FLOPs: 236G, Params: 58.94M, FPS: 20.73). The next best method, UperNet with a Swin-T backbone, achieves 72.34% mIoU, 85.39% pixel Acc, and 83.35% F_1 score. This result highlights the effectiveness of transformer-based architectures in capturing the complex spatial relationships in UAV imagery. Traditional CNN-based models like FCN, PSPNet, and DeepLabV3+ achieve mIoUs around 68-70%, with relatively high FLOPs and modest speed. UperNet with ResNet-50 improves the mIoU to 70.10%, showing that a stronger decoder helps, but at the cost of increased parameters to 64.04M. Mask2Former also performs competitively, particularly in segmenting Barley, where it achieves an IoU of 54.08%. OCRNet with HRNet-w48 offers a good balance of accuracy and efficiency, achieving strong segmentation performance across all categories,

with mIoU values of 70.11%. Segmentation methods utilizing convolutional backbones such as ResNet-50 generally exhibit lower performance compared to transformer-based approaches, particularly in distinguishing Barley from other classes.

The superior performance of Transformer-Based methods suggests that integrating attention mechanisms effectively captures the contextual information in UAV imagery. Traditional CNN-based methods such as FCN, PSPNet, and DeepLabV3+ provide competitive performance but tend to underperform in differentiating between similar crop types.

V. DISCUSSION

This study contributes to two active research areas: learning with noisy labels and semi-supervised semantic segmentation. Compared to prior work, the proposed MulMatch-CL framework builds upon the principles of consistency-based semi-supervised methods [12], [34], while also addressing a key limitation they do not fully resolve: the impact of noisy labels in UAV-based crop images. Although robust learning methods such as Confident Learning have been explored in medical imaging contexts [38], their application to UAV-based semantic segmentation remains limited. This work bridges this gap by integrating these paradigms in a unified framework tailored to the challenges of crop segmentation in UAV-based remote sensing. Evaluated on the Barley Remote Sensing Dataset, MulMatch-CL outperforms robust learning methods, semi-supervised methods, and traditional supervised baselines across all metrics, including mIoU, pixel accuracy, precision, recall, and F_1 score. The ablation study demonstrates that Confident Learning and MulMatch each independently contribute to performance gains, with their integration yielding the most robust results. These improvements validate our hypothesis that jointly addressing label noise and leveraging unlabeled data can lead to more robust and generalizable segmentation models for crop monitoring tasks. Unlike FixMatch, which relies solely on single augmentation streams based on confidence thresholds. The proposed method balances both precision and recall, leading to more accurate segmentation. Moreover, the ablation results show that Swin Transformer is the most effective backbone for the proposed framework, delivering superior segmentation performance while maintaining efficient inference.

While MulMatch-CL demonstrates strong performance on the Barley Remote Sensing Dataset, it is important to note that the dataset covers a relatively small and geographically constrained area. This limitation raises questions about the model's ability to generalize to larger or more diverse agricultural regions. Differences in crop types, growth stages, imaging conditions, and geographic variability may affect performance. Therefore, future evaluations across broader spatial extents and multi-site datasets are necessary to validate the scalability and robustness of the framework in real-world, heterogeneous environments. Additionally, the use of multiple strong augmentation streams improves model generalization but also increases computational overhead during the training process. Despite these limitations, the current findings highlight the promise of the proposed framework for crop segmentation under limited and noisy supervision, laying a solid foundation for future scalable solutions in precision agriculture.

VI. CONCLUSION

This study addresses key challenges in training Transformer models for crop segmentation using UAV remote sensing data, particularly the scarcity of labeled data and the presence of noise in manually annotated datasets. To this end, this study proposes MulMatch-CL, a novel teacher-student framework that integrates the MulMatch framework with Confident Learning. The proposed method achieves 78.42% mIoU, 89.10% pixel Acc, and 87.58% F_1 score on the Barley Remote Sensing Dataset, significantly outperforming traditional supervised baselines, robust learning strategies, and advanced semi-supervised approaches. Ablation studies further confirm that both MulMatch and Confident Learning independently enhance the model performance, with their combination yielding the best performance at 78.42% mIoU. Additionally, the use of multiple strong augmentation streams increases mIoU from 77.18% to 78.42% compared to a single stream version of the proposed method, confirming that diverse augmentations enhance model generalization. By improving the data quality and efficiently leveraging limited labeled data, the proposed method provides a robust solution for crop segmentation.

Future work will focus on extending the applicability and scalability of the proposed approach. Firstly, the framework could be extended to other remote sensing tasks beyond crop segmentation, such as land cover classification or weed detection. Furthermore, although the method demonstrates strong performance on the current dataset, further evaluation across broader geographic regions, diverse crop types, and varied environmental conditions is necessary to assess real-world generalization. Additionally, integrating self-supervised pretraining or domain adaptation techniques could improve performance when transferring to new domains with limited or no labels, making Transformer-based architectures even more effective for large-scale precision agriculture.

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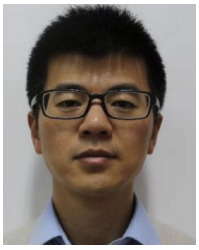

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